

Jamal Al Jazeer F

Biomedical Engineer (BE) — Healthcare Equipment & Service Engineer — AIML Minor

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Nagercoil, Tamil Nadu, India



Professional Summary

Biomedical Engineering graduate with hands-on experience in biomedical equipment maintenance, troubleshooting, preventive maintenance (PPM), and healthcare technology support in hospital environments. Strong exposure to patient monitoring systems, ventilators, infusion devices, and diagnostic equipment. Skilled in biomedical signal processing (ECG, EMG, PPG, EEG), IoT-based healthcare systems, and technical documentation. Seeking entry-level opportunities in Saudi Arabia or UAE within biomedical engineering, medical equipment service, application support, or QA/QC roles in healthcare and medical device industries.

Core Competencies

- **Biomedical Equipment Maintenance:** Troubleshooting, servicing of patient monitors, ventilators, infusion pumps, ECG machines
- **Preventive Maintenance (PPM) & Calibration:** Equipment inspection, performance checks, reliability assurance
- **Medical Device Support:** Installation, commissioning, and technical assistance in hospital environments
- **Quality Control (QC):** Equipment testing, validation, safety checks, compliance support
- **Clinical Engineering:** Hospital biomedical equipment lifecycle support
- **Biomedical Signals:** ECG, EMG, PPG, EEG monitoring and analysis
- **Documentation & Compliance:** Technical reports, maintenance logs, safety protocols
- **IoT & Embedded Systems:** Healthcare device integration and monitoring systems
- **Tools & Programming:** Excel, Power BI, Python (Basic), C++ (Basic)

Internship Experience

Biomedical Engineering Intern

2025 – 2026

G. Kuppuswamy Naidu Memorial (GKNM) Hospital, Coimbatore

- Assisted in maintenance, servicing, and troubleshooting of biomedical equipment including patient monitors, ventilators, infusion pumps, and imaging systems.
- Performed preventive maintenance (PPM) activities and equipment performance checks ensuring operational reliability and patient safety.
- Supported biomedical engineers in calibration observation, technical inspections, and equipment fault diagnosis.
- Documented equipment servicing procedures and assisted in healthcare equipment lifecycle management.

Academic Projects

Wireless PPG-Based IoT Blood Pressure Monitoring System

- Designed system for non-invasive blood pressure monitoring using PPG sensor data and IoT communication.
- Worked on signal acquisition, wireless transmission, and physiological data interpretation.

EMG-Controlled Rehabilitation Therapy System

- Developed EMG-based control system for rehabilitation applications in stroke and neurological therapy.
- Gained exposure to biomedical signal processing and assistive healthcare technology design.

Bone Anchored Hearing Aid Prototype

- Developed low-cost biomedical prototype using bone conduction technology.
- Participated in hardware integration and biomedical device design process.

MedVault – Medical Record Management System

- Designed secure healthcare data management system for structured medical records handling.
- Improved accessibility and data organization in digital healthcare systems.

Education

B.E. Biomedical Engineering (AIML Minor)

2022 – 2026

Hindusthan College of Engineering and Technology, Coimbatore

Publications

- Wearable Non-Invasive Optical Monitoring for Bladder Dysfunction and Enuresis (DOI: 10.63001/tbs.2024.v19.i03.pp1125)
- EMG-Based Game Therapy for Stroke and Cerebral Palsy Rehabilitation (ISBN: 978-93-342-4986-6)

Additional Information

- Willing to relocate immediately to Saudi Arabia / UAE
- Languages: English, Tamil, Malayalam
- Nationality: Indian